



Operations Research

Prof. Dr. Martin Bichler,
Halil Bayrak, Janik Bürgermeister, Teodora Dobos,
Jan Höhener, Abheek Ghosh

Exercise Sheet 7

Training Exercises

Exercise T7.1. *Cutting*

A carpentry shop receives an order from a customer to produce 30 boards of length $5m$ (type A), 10 boards of length $10m$ (type B) and 13 boards of length $12m$ (type C). She has $30m$ length boards available for cutting. She incurs a cost of 1 Euro per board used. Due to technical limitations, additional costs are incurred in the following cases:

- If the cutting pattern contains more than 2 boards of length $5m$.
- If the cutting pattern contains more than 1 board of length $10m$.

For each case that applies, there is an additional cost of 1 Euro. The carpentry would like to solve the problem by means of column generation.

- a) For the CG algorithm, an optimal cutting pattern must be generated in each iteration with respect to a given utility vector $u = (u_A, u_B, u_C) \in \mathbb{R}^3$ and the costs described above. Let u_T denote the utility of a board of type $T \in \{A, B, C\}$. Formulate an IP that solves the problem "Find an optimal cutting pattern with respect to u " and implement it in Gurobi.
- b) Denote by x_1, x_2, x_3 the initial basis, where x_1 is the pattern with six $5m$ boards, x_2 is that with three $10m$ boards, and x_3 is that with two $12m$ boards. Indicate which basis you get after 2 iterations of the CG algorithm.

Exercise T7.2. *Hiking trip*

Joe Hännies is organizing a hike with his friends. He wants to take bread with different spreads with him. There are a total of $m \in \mathbb{N}$ different spreads and he wants to take at least b_i loaves with spread i with him. He has identical boxes at his disposal for transportation. As Joe Hännies is a gourmet, he pays attention to how well spreads harmonize aromatically. For the spread i , let $H_i \subseteq \{1, \dots, m\}$ be the quantity of spreads that harmonize with i . Then a filling pattern of a box is only *valid*, if the following condition is met for all spreads i : The number of loaves with i harmonizing spreads is at least as large as the number of loaves with spread i . Furthermore each box can hold a maximum of G loaves and no box may remain empty.

For example, let's assume $H_1 = \{5, 8\}$ and fill a box with two loaves of bread with spread 1.

Then, according to the above condition, at least two more loaves with spread 5 or 8 would have to be added to the box.

Joe Hännies wants to take as few boxes as possible with him and wants to solve this problem using column generation. To do this, he first sets up the following IP:

$$\begin{aligned} \min \quad & \sum_{i=1}^n c_i x_i \\ \text{s.t.} \quad & Ax = b \\ & x \in \mathbb{N}_0^n. \end{aligned}$$

The matrix A has m rows and n columns, where each column describes an admissible filling pattern of a box. Furthermore, $c = (1, \dots, 1)^T \in \mathbb{R}^n$. The column $a \in \mathbb{N}_0^m$ therefore describes a box in which there are exactly a_i loaves with spread i . Furthermore, x_i indicates how often the filling pattern in the i -th column is used.

- a) Describe the condition “ $a = (a_1, \dots, a_m)^T \in \mathbb{N}_0^m$ is valid” by linear (in-)equalities.
- b) In this subtask we assume that $G = 3$ and $m = 2k$ for $k \in \mathbb{N}$ is arbitrary. Specify the number of valid filling patterns for each of the following two cases:
 - Case 1: $H_i = \{i\}$ for all $i \in \{1, \dots, m\}$
 - Case 2: $H_i = \begin{cases} \{i+1\}, & \text{for } i \text{ odd} \\ \{i-1\}, & \text{for } i \text{ even} \end{cases}$
- c) Explain in which of the two cases the use of Column Generation is particularly suitable?
- d) For this subtask, we assume that $m = 4$, $G = 4$ and $H_1 = \{1\}$, $H_2 = \{2\}$, $H_3 = \{2, 4\}$, $H_4 = \{4\}$. Specify the pricing problem and an optimal solution for the first iteration of the column generation algorithm (the outgoing base variable does not have to be determined). Start with the following base:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Self-study Exercises

Exercise S7.1. *Cutting*

A carpentry shop receives an order from a customer to produce 16 boards of length $4m$, 7 boards of length $7m$, and 13 boards of length $9m$. Fabienne is responsible for the order and has $19m$ length boards available for cutting. She incurs a cost of 1 Euro per board used. Her colleagues tell her to consider the following cutting patterns, where $a_1 = (a_{11}, a_{12}, a_{13})^T = (3, 0, 0)^T$ contains three boards of length $4m$, $a_2 = (a_{21}, a_{22}, a_{23})^T = (0, 2, 0)^T$ contains two boards of length $7m$, and $a_3 = (a_{31}, a_{32}, a_{33})^T = (0, 0, 1)^T$ contains one board of length $9m$. Let furthermore denote with $x_1, x_2, x_3 \in \mathbb{N}_{\geq 0}$ how often each cutting pattern is used. This results in the IP:

$$\begin{aligned} (M) \quad & \min x_1 + x_2 + x_3 \\ & s.t. \quad 3x_1 \geq 16 \\ & \quad \quad 2x_2 \geq 7 \\ & \quad \quad x_3 \geq 13 \\ & \quad \quad x_1, x_2, x_3 \in \mathbb{N}_{\geq 0} \end{aligned}$$

- a) Fabienne thinks the cutting patterns a_1, a_2, a_3 are suboptimal. How and why can she use the column generation algorithm to show her colleagues that she is right?
- b) Specify the pricing problem and an optimal solution for the first iteration of the column generation algorithm (the outgoing base variable does not have to be determined). Start with the base a_1, a_2, a_3 .
- c) Using your results from part b), state the adapted version M' of M with the newly generated column.
Note: You may use placeholder variables if you did not solve part b).

Exercise S7.2. *Bin Packing*

We consider the bin packing problem: Given n items $i \in N = \{1, \dots, n\}$ with weights g_i . These items are to be distributed among bins, with items of total weight G fitting into each bin. The number of bins used is to be minimized.

- For a given instance of the bin packing problem, the columns of the matrix $A \in \{0, 1\}^{n \times \alpha}$ encode all α filling patterns of a bin. If a is a column of A , then $a_i = 1$ exactly when there is item i in the corresponding pattern in the bin. Formulate the bin packing problem using A as IP.
- Now let $n = 10$ objects be given, where $g_i = i$. Each bin can further hold a weight of $G = 10$. Perform an iteration of the CG algorithm applied to the LP relaxation of the bin packing problem. Start with the base where each item goes into its own bin.